Astro 142 Discussion 3

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Short Answer Questions

Describe the difference between a class and an object

A class defines the structure and behavior (attributes and methods), and does not occupy memory itself.
A concrete example created from a class, and has values stored in memory

Discussion Activity

Discussion 3 Activity

This script demonstrates object-oriented programming using astrophysical examples. It defines classes for stars, main sequence stars, black holes, and binary systems, with methods to calculate properties like luminosity, surface gravity, and orbital periods. In the demo, it creates example objects (a star, a black hole, and a binary system) and prints their physical characteristics.

```python
# ============================================================
# Class: Black Hole (Schwarzschild)
# ============================================================
class BlackHole:
    """Simple Schwarzschild black hole with name, mass, and radius."""
    G = Star.G
    c = 2.99792458e8
    M_SUN = Star.M_SUN

    def __init__(self, name, mass):
        self.name = name
        self.mass = mass  # solar masses
        self.radius = self.schwarzschild_radius()

    def schwarzschild_radius(self):
        return (2 * BlackHole.G * self.mass * BlackHole.M_SUN) / (BlackHole.c**2)

    def summary(self):
        print(f"Name: {self.name}")
        print(f"  Type         = Black Hole")
        print(f"  Mass         = {self.mass:.3f} M☉")
        print(f"  R_s          = {self.radius:.3e} m")

# ============================================================
# Demo
# ============================================================
if __name__ == "__main__":
    print("\n=== Example: Main Sequence ===")
    sun = MainSequenceStar("Sun", 1.0)
    sun.summary()

    print("\n=== Example: New Star ===")
    newstar = MainSequenceStar("Alpha", 2.0)
    newstar.summary()

    print("\n=== Example: Black Hole ===")
    bh = BlackHole("Ton 618", 66*(10**9))
    bh.summary()

    print("\n=== Example: Binary System ===")
    binary = BinaryStar(newstar, bh, separation=5.0)
    binary.summary()
```

Using classes organizes related data and behavior, making the code modular, reusable, and easy to extend, such as modeling complex systems like binary stars. It also improves readability and maintainability.

However, it adds extra code and complexity, which can be unnecessary for small tasks, and inheritance can make the design harder to manage if overused.